

Motivation

This project aims to analyze how consumer spending behavior changes during large promotional events such as Black Friday. I became interested in this topic after previously working on a presentation related to marketing strategies and consumer behavior. Discounts and pricing strategies are widely used in e-commerce platforms, and I wanted to explore how these factors may influence purchasing decisions using real-world data and data science techniques. The project also provides an opportunity to apply statistical analysis and machine learning methods to a large-scale consumer dataset.

Data Source

The primary dataset used in this project is the Black Friday dataset obtained from Kaggle. The dataset contains customer demographic information such as age, gender, occupation, marital status, and city category, along with product categories and purchase amounts.

Since the main dataset does not contain direct discount information, an additional e-commerce pricing dataset from Kaggle was incorporated as a supplementary data source. This supplementary dataset includes category-level product price and discount percentage information. Product categories were used as proxy variables to provide indirect discount-related insights and strengthen the overall analysis.

Data Analysis

The analysis began with data cleaning and preprocessing, including handling missing values and preparing categorical variables for analysis. Exploratory Data Analysis (EDA) was conducted to examine the structure of the dataset, distributions of purchase amounts, and relationships between demographic variables and spending behavior.

Several statistical techniques were applied during the analysis. Hypothesis testing methods such as t-tests and ANOVA were used to investigate whether statistically significant differences existed between different demographic groups and product categories.

To enrich the analysis, summary statistics from the supplementary e-commerce pricing dataset were examined and compared with product categories in the Black Friday dataset. This enrichment process provided indirect insights into possible discount-related purchasing patterns.

In the machine learning stage, a Random Forest Regressor model was implemented to predict purchase amounts based on demographic and product-related variables. Model performance was evaluated using Mean Absolute Error (MAE) and R^2 score metrics. Feature importance analysis was also conducted to determine the most influential variables affecting purchase behavior.

Additionally, the Random Forest model was compared with a Linear Regression model. The comparison showed that Random Forest achieved significantly better predictive performance, suggesting that nonlinear relationships exist between the variables and purchase amounts.

Findings

The analysis demonstrates that consumer spending behavior is strongly influenced by product categories and demographic characteristics. Statistical tests showed significant differences in purchase behavior across groups.

The machine learning results revealed that product categories are the strongest predictors of purchase amounts. The Random Forest Regressor achieved an R^2 score of approximately 0.64,

while the Linear Regression model achieved a substantially lower score of approximately 0.16. This indicates that the relationships between variables and purchase behavior are not purely linear.

The enriched pricing and discount information also suggested that pricing strategies and discount-related patterns may influence purchasing behavior, even though direct discount data was not available in the primary dataset.

Limitations and Future Work

One important limitation of this study is that the primary dataset does not directly include discount information. Therefore, product categories and supplementary pricing data were used as proxy variables for discount-related behavior, which may not fully capture real discount effects.

Future studies could incorporate datasets containing direct product-level discount information and customer transaction histories. Additional machine learning models such as XGBoost or Neural Networks could also be explored and compared to improve predictive performance. Furthermore, more advanced feature engineering and dataset merging techniques could strengthen the analysis and provide deeper insights into consumer purchasing behavior.